

KAMRAN ALAM

Undergraduate Researcher · Software Engineering · AI Systems & Memory Management

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RESEARCH INTERESTS

AI safety evaluation methodologies, adaptive systems behavior, simulation-based analysis of algorithmic performance, and the intersection of OS-level memory management with AI workload optimization. Interested in contributing to empirical evaluations of AI system robustness, particularly in the context of biological and technical risk.

PUBLICATIONS

Peer-Reviewed Journal Article

April 2026

Static Versus Adaptive Paging: A Simulation-Based Comparison of Windows-Style and Linux Memory Management Under Realistic Workloads

Kamran Alam, Nimra Babar, Seema Sultana Bhurgri, Abdul Qayoom, Asif Irshad, Muhammad Ameen Chhajro

Annual Methodological Archive Research Review (AMARR), Vol. 4, No. 4, pp. 121–144, 2026.

DOI: <https://doi.org/10.5281/zenodo.19878166> | URL: <https://amresearchjournal.com/index.php/Journal/article/view/1951>

Key Contributions:

- Designed and implemented a Java-based simulation comparing FIFO, LRU, and Optimal page replacement algorithms across four realistic workload patterns (Random, Browser, Sequential File, Video Streaming).
- Developed a theoretical model of Linux's adaptive Active/Inactive two-list paging architecture and contrasted it quantitatively against static Windows-style policies.
- Identified that adaptive policies reduce page faults by up to 30% under sequential file access vs. rigid static approaches (FIFO/LRU), with direct implications for AI/ML workload memory optimization.
- Paper addresses a research gap: most prior work used purely random reference strings rather than realistic workloads — our simulation bridges this gap.

RESEARCH & TECHNICAL PROJECTS

NDMA/PDMA Disaster Data Pipeline

2026

Data Science Research Project · SMIU

- Engineered an end-to-end Python pipeline to scrape, download, and convert government disaster PDFs (NDMA/PDMA) into structured Excel datasets using Tesseract OCR and pandas, forming the data foundation for a multi-stage disaster intelligence system.
- Constructed a hierarchical data schema to categorize disaster events (floods, monsoons, thunderstorms) by region, severity, and date — enabling systematic clustering analysis to identify high-risk geographic zones across Pakistan.
- Ongoing: developing a predictive model to forecast future disaster-prone locations and estimate required relief expenditure per region, supporting data-driven emergency preparedness planning.

EDUCATION

BS Software Engineering

Jan 2023 – Jan 2027 (Expected)

Sindh Madressatul Islam University (SMIU), Karachi, Pakistan

CGPA: 3.16 | Currently 6th Semester

Relevant coursework: Data Science, Computer Architecture & Networking, Software Project Management, Software Re-Engineering, Web Technology, Operating Systems.

Intermediate — Computer Science

Oct 2021 – Dec 2023

Superior College, Karachi

Modern Web & App Development Course

Mar 2025

Saylani Mass IT Training (SMIT), Karachi

TECHNICAL SKILLS

Research Methods:	Simulation design, algorithm performance analysis, theoretical modeling, workload benchmarking
Programming:	Python, Java, Go, C, C++, C#, Assembly (x86-64)
AI/Data Science:	pandas, NumPy, Tesseract OCR, data extraction & cleaning, hierarchical schema design, clustering analysis, predictive modeling (in progress), structured dataset construction
Web & Systems:	HTML, CSS, JavaScript, React, TypeScript, Node.js, PostgreSQL, MongoDB
Tools:	Git/GitHub, GitHub Actions, Linux, VS Code, Jupyter Notebook

RELEVANT EXPERIENCE

Web Development Team Member

Sep 2025 – Dec 2025

Ziraal | (Non-Profit Organization) Karachi, Pakistan

- Developed and maintained web features for a non-profit organization, contributing to their digital outreach and community service delivery platform.
- Collaborated in a team environment practicing code review and agile-style project management.

Freelance Web Developer & Graphic Designer

Jan 2021 – Present

Fiverr | Remote

- 4+ years independently scoping, delivering, and iterating on technical projects for international clients — demonstrating initiative and self-directed work capacity relevant to independent research.

Home Tutor — Mathematics

Oct 2021 – Present

Self-employed | Karachi, Pakistan

- Teaching Matric-level mathematics for 3+ years; develops ability to explain complex concepts clearly — directly relevant to research communication and mentorship contexts.

CERTIFICATIONS & TRAINING

- Complete HTML + CSS – Build Responsive Website (Udemy)
- Mastering x86-64 Real Assembly Language / ASMCert+ (Udemy)
- WordPress Development (Udemy)
- Effective Leadership (Online)
- National Certificate of Vocational Qualification – IT, Computer Operator, NVQF Level 2 (NAV TTC)

LANGUAGES

Urdu (Native) English (Fluent) Pashto (Native)